

Analysis and Comparison of Current Research in Blockchain Security

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Abstract

Blockchain technology enables decentralized agreement and tamper-resistant ledgers. Despite strong cryptographic foundations, blockchains remain vulnerable to attacks that exploit incentives, network dynamics, and adversarial behaviors. This report analyzes and compares three scientific papers addressing complementary aspects of blockchain security: [1] adversarial Ethereum transaction detection using Generative Adversarial Networks (GANs) and deep learning, [2] a mining attack that leverages partial block disclosure and zero-knowledge proofs to attract rational miners, and [3] the foundational selfish mining result showing that majority honesty is not sufficient for incentive compatibility. For each paper, we summarize objectives, methodology, and results, then critically assess strengths and limitations. We then compare contributions, identify trends, challenges, and opportunities, and propose open research questions and future directions.

Introduction

Blockchains are distributed systems designed to operate without a central authority. They maintain a shared ledger through consensus, often under adversarial conditions. In Proof-of-Work (PoW) systems, miners compete to append blocks, and rewards incentivize participation. However, security is not guaranteed solely by cryptography. In practice, security also depends on assumptions about rational behavior, network latency, and how participants react to incentives.

A common belief is that PoW blockchains are secure if honest miners control the majority of hash power. Yet research shows that a minority can deviate from the protocol and still gain disproportionate revenue. Such incentive attacks threaten decentralization by encouraging miners to join profitable deviant pools. At the same time, blockchains generate large observable datasets (transactions, events, logs) that enable data-driven security analytics. Machine learning can help detect fraud and anomalies, but it can also be attacked through adversarial examples and distribution shifts.

This report studies three papers that jointly illustrate these issues from different layers. The first paper proposes a threat hunting pipeline for Ethereum that uses a conditional GAN to generate realistic fake transactions and a Bi-LSTM to detect adversarial transactions. The second paper introduces Partial Selfish Mining (PSM) and Advanced PSM (A-PSM), where attackers share partial block data and use zero-knowledge proofs (ZKPs) to attract rational miners. The third paper establishes the selfish mining strategy and proves that Bitcoin mining is not incentive-compatible.

1 Research Papers Explanation

1.1 Paper 1: Generative Adversarial Networks for Cyber Threat Hunting in Ethereum Blockchain

1.1.1 Objectives

The primary objective of the paper is to design an intelligent cyber threat hunting model capable of detecting adversarial and zero-day attacks in the Ethereum blockchain. The authors aim to overcome the limitations of traditional machine learning-based intrusion detection systems, which often fail to detect highly realistic and previously unseen attack patterns.

1.1.2 Methodology

The proposed approach is based on a two-phase deep learning framework.

In the first phase, the authors employ a Conditional GAN for tabular data (CTGAN) to generate realistic fake Ethereum transactions from genuine transaction data. The GAN consists of two deep neural networks: (i) a *generator* that learns to produce synthetic transactions from a noise vector and conditional column labels, and (ii) a *discriminator* that learns to distinguish real transactions from generated ones and provide feedback to improve the generator. Training relies on the Wasserstein GAN (WGAN) loss function to improve stability and the quality of generated samples.

In the second phase, real and generated transactions are combined and used to train a bi-directional Long Short-Term Memory (bi-LSTM) network to identify adversarial transactions. The bi-LSTM analyzes transaction sequences in both forward and backward directions to capture temporal dependencies. From a security perspective, the authors interpret this detection process using the Cyber Kill Chain (CKC) model and argue that identifying adversarial transactions at the *delivery* stage enables early interruption of the attack before later stages such as exploitation and command and control. Feature selection is performed using the ExtraTreesClassifier, and data normalization is applied prior to training.

1.1.3 Results

The experimental evaluation demonstrates that the GAN model is able to generate highly realistic transactions, achieving a similarity score of approximately 82.51% compared to real Ethereum transactions. The bi-LSTM detection model achieves excellent performance, with a reported accuracy of up to 99.98% in identifying adversarial transactions.

These results indicate that the proposed model is highly effective in detecting sophisticated and previously unknown attacks in blockchain environments.

1.1.4 Strengths

The main strengths of the proposed approach are:

- Innovative use of GANs to simulate adversarial attacks in Ethereum blockchain systems.
- High detection accuracy against adversarial and zero-day attacks.
- Combination of data generation and sequence-based deep learning for improved robustness.
- Use of real and publicly available Ethereum transaction data.

1.1.5 Limitations

Despite its strong performance, the proposed model has several limitations:

- High computational and energy costs due to GAN training.
- Dependence on the quality and integrity of the training dataset.
- Limited evaluation in real-time or large-scale production blockchain environments.

1.2 Paper 2: Mining Attack with Zero Knowledge in the Blockchain

1.2.1 Objectives

Mining attacks involve a group of miners who do not follow the protocol rules in order to gain more rewards. They are considered a serious threat to Proof-of-Work (PoW) blockchain systems. Selfish mining is an attack where a miner withholds valid blocks and releases them strategically to create forks, wasting honest miners' work and increasing the attacker's rewards. This paper introduces a new mining attack that leverages zero-knowledge proofs. Two versions of this attack are proposed: Partial Selfish Mining (PSM) and Advanced Partial Selfish Mining (A-PSM), the objective of this is to show how selfish mining attacks has been improved and also to demonstrate its key limitation

and how this things can increase attacker profitability under certain mining power and work conditions,also to Show how zero-knowledge proofs can be used to credibly attract rational miners without revealing full block information.

1.2.2 Methodology

The authors show how each type of attack works. First, they define the classic selfish mining attack as a scenario in which the attacker finds a valid block but keeps it private and start mining on top of it so when honest miners find and publish their block the attacker strategically releases their hidden blocks which causes two competing chains , due to the network propagation honest works get perturbed which earns for the attacker more block reward ,However, this attack is rarely observed in practice cause miners don't trust blindly and they also require proof of block validity. So, researchers tried to improve selfish mining by proposing algorithms for optimal selfish strategies and block withholding techniques, but all of these approaches still depend heavily on the attacker's mining power. This is where PSM comes in: instead of mining alone, the attacker attracts and cooperates with rational miners to increase their effective mining power, but how those PSM and A-PSM work ?

Differ to the classic mining attack ,PSM attackers reveal only part of the block and hide critical information this partial information look real but doesn't give full control to others ,this works because it attracts rational miners where they help increase the attacker power and their chance of winning the next block race , as for A-PSM it is a same attack but more advanced because the attacker can constantly monitor the public chain and release blocks at the best possible moments. To analyze these strategies, the paper formally models the mining process by defining different types of miners (attacker, rational miners, and public miners, and attracted miners) and representing the attack dynamics using state machines. It then uses mathematical formulations, including expected reward and Relative Extra Reward (RER), to characterize the incentives of different strategies, and introduces cryptographic mechanisms such as zero-knowledge proofs (ZKP) and smart-contract-based deposits to ensure trust between the attacker and the rational miners. Finally, Monte Carlo tables are employed to validate the analytical models.

1.2.3 Results

The results of the paper show that classic selfish mining becomes effective only when the attacker controls more than 33% of the mining power, which limits its practicality. In contrast, the analysis demonstrates that Partial Selfish Mining significantly lowers this requirement by attracting rational miners. Using analytical formulas, the authors show that it is profitable for rational miners to join the attacker's private branch rather than mine honestly, as long as their mining power is smaller than the attacker's. This

incentive is quantified using the Relative Extra Reward (RER), which remains positive across a wide range of attacker power and network conditions. As results this maPSM more profitable than classic selfish mining for small to moderate attacker power. Similarly, Advanced Partial Selfish Mining (A-PSM) further improves this advantage by adapting block release strategies, enabling both the attacker and rational miners to achieve higher rewards than with honest or selfish mining, and making the attack stable and practical in real-world conditions.

1.2.4 Strengths

- Introduces a realistic extension of selfish mining using economic and cryptographic insights.
- Validates the model through mathematical analysis and Monte Carlo simulations.
- Considers practical feasibility and trust mechanisms in PoW blockchains.

1.2.5 Limitations

- Assumes rational miner behavior and simplified network conditions.
- Requires additional cryptographic and smart-contract infrastructure.
- Focuses mainly on Proof-of-Work systems.

1.3 Paper 3: Majority Is Not Enough: Bitcoin Mining Is Vulnerable

1.3.1 Objectives

This paper aims to challenge the common belief that Bitcoin mining is secure as long as no single miner or mining pool controls more than 50% of the network’s computational power. The authors argue that the Bitcoin protocol is not incentive-compatible, meaning that rational miners may benefit from deviating from the honest mining strategy even when they control a minority of the mining power. The primary objective is to demonstrate that a group of colluding miners can gain unfair economic advantages through a strategy called *Selfish Mining*. The paper also seeks to analyze the consequences of this behavior on the decentralization of Bitcoin and to propose a protocol-level solution that mitigates this vulnerability.

1.3.2 Methodology

The authors adopt a theoretical and analytical methodology, complemented by simulations. They model Bitcoin miners as rational agents who aim to maximize their revenue.

Mining power is represented as a fraction of the total network power, and block discovery is modeled probabilistically.

The paper introduces the Selfish Mining strategy, in which a mining pool withholds newly mined blocks instead of publishing them immediately. By selectively releasing these blocks, selfish miners can force honest miners to waste computational resources on blocks that will not be included in the main blockchain.

To evaluate this strategy, the authors construct a state-machine model that represents the difference in length between the private chain of selfish miners and the public chain mined by honest participants. Using this model, they derive mathematical expressions for expected revenues under different conditions, including network propagation effects captured by the parameter γ .

Finally, the analytical results are validated through Monte Carlo simulations, which confirm the theoretical findings.

1.3.3 Results

The results show that selfish mining can be profitable even for minority mining pools. Contrary to the widely assumed 50% security threshold, the authors demonstrate that selfish mining becomes advantageous with as little as 25% to 33% of the total mining power, depending on network conditions.

The study also reveals that once a selfish mining pool exceeds this threshold, its revenue increases faster than its size, encouraging rational miners to join the pool. This dynamic can lead to rapid centralization and ultimately threaten Bitcoin's decentralized nature.

Additionally, the authors show that under the current Bitcoin protocol, the attack threshold can be very close to zero if selfish miners control block propagation effectively. To address this issue, they propose a simple modification to the protocol: when miners encounter chains of equal length, they should propagate all of them and randomly choose which one to mine on. This change raises the attack threshold to 25% and improves network security without requiring a hard fork.

1.3.4 Strengths

- Uses a rigorous analytical framework combining game theory, probability theory, and simulations to reveal a critical vulnerability in Bitcoin's mining protocol.
- Considers miners as rational, profit-maximizing agents, which better reflects real economic behavior than assuming honest participation.
- Provides a high-impact contribution by identifying a realistic attack that does not rely on cryptographic weaknesses but on incentive misalignment.

1.3.5 Limitations

- Relies on simplified network assumptions, particularly regarding block propagation and communication delays, which may differ in real-world networks.
- Assumes fully rational miner behavior, while in practice some miners may act altruistically or prioritize network stability.
- The proposed protocol modification mitigates the attack but does not completely eliminate the risk of mining centralization.

2 Comparison and Contributions

2.1 How the papers differ

The papers address blockchain security at different layers.

Paper [1] focuses on the data and transaction layer, proposing a detection pipeline. Paper [2] focuses on protocol-level strategic behavior and introduces a new attack mechanism that exploits ZKPs. Paper [3] focuses on incentive compatibility in Bitcoin and provides the foundational selfish mining model.

Their methods differ as well. Paper [1] uses machine learning (GAN + Bi-LSTM) and reports detection accuracy. Papers [2] and [3] use game-theoretic revenue modeling and analyze when deviation is profitable.

2.2 How the papers complement each other

Paper [3] establishes that incentive misalignment can break decentralization even without majority control. Paper [2] extends this by demonstrating that new cryptographic tools can expand attack strategies and attract rational miners through verifiable signaling. Paper [1] shows that, even beyond mining, adversarial behaviors at the transaction layer require robust monitoring and detection, especially when attackers craft inputs to evade ML-based defenses.

Together, these papers suggest that blockchain security must be multi-layer: protocol rules should reduce the profit of deviations, and monitoring systems should detect emerging attacks and anomalous behavior.

3 Trends, Challenges, and Opportunities

3.1 Trends

A key trend is the integration of machine learning into blockchain security monitoring. As blockchain usage grows, manual analysis becomes infeasible, motivating automated analytics. Another trend is the expansion of mining attack strategy spaces. Researchers increasingly explore timing, partial disclosure, and multi-agent dynamics rather than simple deviations. A third trend is the interaction between cryptographic tools and incentives. Privacy and verifiability tools can reshape strategic behavior.

3.2 Challenges

A core challenge is incentive alignment: preventing profitable deviations without centralizing control. Another challenge is realism in evaluation. Results may depend on network parameters, miner behavior assumptions, and dataset representativeness. For ML-based approaches, adversarial robustness and distribution shift are persistent challenges. Detectors must work under evolving conditions and adaptive adversaries.

3.3 Opportunities

Opportunities include hybrid systems combining protocol-level defenses and real-time detection. Improving tie-breaking and propagation mechanisms may reduce selfish mining incentives. Designing ZK integrations with incentive analysis may prevent unintended collusion benefits. For ML, building benchmarks with realistic adversarial scenarios and reporting richer metrics can improve trust in results.

4 Open Research Questions and Future Directions

4.1 Open Research Questions:

The following open research questions arise from our research papers:

4.1.1 Paper 1: Generative Adversarial Networks for Cyber Threat Hunting in Ethereum Blockchain

- How robust is the proposed CTGAN and bi-LSTM framework when deployed in real-time Ethereum environments with continuously evolving transaction patterns and adversarial strategies?

- To what extent can the model generalize to other blockchain platforms or different types of adversarial behaviors without requiring extensive retraining or additional labeled data?

4.1.2 Paper 2: Mining Attack with Zero Knowledge in the Blockchain

- How can Proof-of-Work blockchains be redesigned or secured to prevent incentive-based mining attacks such as PSM and A-PSM without reducing decentralization?
- Could advanced cryptographic techniques, such as zero-knowledge proofs, unintentionally introduce new security risks in blockchain systems?

4.1.3 Paper 3: Majority Is Not Enough: Bitcoin Mining Is Vulnerable

- How can blockchain consensus protocols be redesigned to ensure incentive compatibility while preserving decentralization, especially in environments with rational and economically motivated participants?
- To what extent can alternative consensus mechanisms or hybrid approaches reduce or eliminate vulnerabilities such as selfish mining without introducing new security or centralization risks?

4.2 Future Perspectives

Taken together, the three analyzed papers highlight a clear evolution of blockchain threats from protocol-level incentive attacks to highly sophisticated, learning-based and cryptography-enhanced strategies. Future research should therefore move toward unified defense frameworks that jointly consider adversarial transaction generation, strategic mining behavior, and collusion mechanisms enabled by advanced cryptographic tools such as zero-knowledge proofs. In particular, integrating machine learning-based threat hunting models with economic and game-theoretic security analyses could improve the early detection of both transaction-level adversarial attacks and mining-level strategic deviations. Additionally, extending these approaches to real-time, multi-blockchain environments and evaluating their robustness against adaptive attackers remains a critical challenge for practical deployment.

5 Conclusion

This report analyzed and compared three papers covering complementary aspects of blockchain security. The Ethereum GAN-based approach shows how AI can strengthen threat hunting by generating realistic adversarial transactions and training deep detectors.

The mining attack with zero knowledge demonstrates that ZKPs can enable new collusion strategies through partial disclosure and credible signaling, expanding the mining attack space. The selfish mining paper provides the foundational insight that PoW mining may not be incentive-compatible and that minority pools can gain extra rewards and drive centralization. Across these works, the main lesson is that blockchain security is not purely cryptographic. It also depends on incentives, network behavior, and adversarial adaptation. Future work should integrate protocol redesign with robust monitoring, develop richer evaluation metrics, and analyze new cryptographic features under realistic incentive models.

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